Antiderivatives and differential equations

Lecture and worked examples

Maths Explorer

Objectives

Calculate antiderivatives; solve a differential equation with an initial condition; interpret an Euler approximation.

Antiderivatives on an interval

At this stage we accept that every continuous function on an interval has antiderivatives; the link with area will be established in chapter 8. An antiderivative of f on an interval I is a differentiable function F such that $F' = f$ on I. If F is an antiderivative, all the antiderivatives on I are $F + C$, $C \in \mathbb{R}$.

Antiderivatives on an interval

If $F' = G' = f$, then $(F - G)' = 0$ on the interval; $F - G$ is therefore constant. Conversely, $(F + C)' = f$. The interval hypothesis is essential: on a union of disjoint intervals, such as $\mathbb{R} \setminus \{0\}$, a union of two intervals, the constant may differ from one interval to the other.

Antiderivatives on an interval

$f(x)$	$F(x)$
$x^n \ (n \in \mathbb{Z},\ n \neq -1)$	$\dfrac{x^{n+1}}{n+1}$
$\dfrac{1}{x}$	$\ln \lvert x \rvert$ sur un intervalle évitant 0
$\dfrac{1}{\sqrt{x}}$	$2\sqrt{x} \quad (x > 0)$
e^x	e^x
$\cos x$	$\sin x$
$\sin x$	$-\cos x$

The additive constant can always be added. The last two lines will be revisited after the study of trigonometry.

Recognising a composition

For a differentiable function u, the antiderivatives of $u'u^n$, $u'e^u$, $\dfrac{u'}{u}$ are respectively $\dfrac{u^{n+1}}{n+1}$ $(n \neq -1)$, e^u, $\ln|u|$, on the intervals where the expressions make sense. Linearity allows us to handle sums.

Recognising a composition

An antiderivative of $6x(x^2+1)^2$ is $(x^2+1)^3$; an antiderivative of e^{3x} is $\dfrac{e^{3x}}{3}$. To impose $F(0)=2$ in the latter case, take $F(x)=\dfrac{e^{3x}}{3}+\dfrac{5}{3}$.

Differential equations

A differential equation relates an unknown function to its derivatives. A solution must be verified on a specified interval.

Differential equations

$y' = ay$ has solutions $y(x) = Ce^{ax}$. For $a \neq 0$, $y' = ay + b$ has solutions

$$y(x) = Ce^{ax} - \frac{b}{a}.$$

If $a = 0$, the equation becomes $y' = b$ and $y(x) = bx + C$. If g is a known particular solution of $y' = ay + f(x)$, the solutions are $y = g + Ce^{ax}$.

Differential equations

If y is a solution, $z(x) = e^{-ax}y(x)$ satisfies $z' = e^{-ax}(y' - ay) = 0$; $z = C$. Conversely, the derivative of Ce^{ax} is aCe^{ax}. For $y' = ay + b$, subtracting the constant solution $-\frac{b}{a}$ reduces it to the homogeneous equation. For a variable right-hand side, do the same with g.

Differential equations

$y' = -2y + 6$ and $y(0) = 1$ give $y = 3 + Ce^{-2x}$, then $C = -2$. The solution $3 - 2e^{-2x}$ increases towards the equilibrium 3 as $x \to +\infty$.

Euler's method

For $y' = ay + b$, with step $h > 0$, we approximate $y(t_k)$ by Y_k according to $t_{k+1} = t_k + h$, $Y_{k+1} = Y_k + h(aY_k + b)$. This is an approximation using the tangent, not an equality for the exact solution. A smaller step may improve the approximation; check the error numerically on a known solution.

Research activity

Cooling of a drink - 40 min.

A drink initially at 80 °C is placed in a room kept at 20 °C. For $t \geq 0$, let $T(t)$ denote its temperature in degrees Celsius, where t is the time elapsed in minutes. We accept the model $T'(t) = -0.1(T(t) - 20)$ and $T(0) = 80$.

Research activity (2)

1. Solve this initial value problem and give the expression of $T(t)$.
2. Study the direction of variation of T and find its limit at $+\infty$. Interpret the results.
3. Find the first time at which the temperature equals $40\,^{\circ}\mathrm{C}$. Give the exact value, then an approximate value in minutes.

Research activity (3)

4. Apply two steps of Euler's method with a step of 1 minute. Compare the approximation obtained after two minutes with the exact value $T(2)$.
5. Give a concrete situation in which the assumption of a constant room temperature would no longer be appropriate.